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Antenna structure and electronic device including same

Abstract

The disclosure relates to a fifth generation (5G) or a sixth generation (6G) communication system for supporting a higher data transfer rate. A dual polarization patch antenna structure and an electronic device including same are provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is based on and claims priority under 35 U.S.C. § 119(a) of a Korean patent application number 10-2022-0094741, filed on Jul. 29, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The disclosure relates to an antenna structure. More particularly, the disclosure relates to a dual polarization patch antenna structure and an electronic device including same.

2. Description of Related Art

(3) Fifth generation (5G) mobile communication technologies define broad frequency bands such that high transmission rates and new services are possible, and can be implemented not only in “Sub 6 gigahertz (GHz)” bands such as 3.5 GHz, but also in “Above 6 GHz” bands referred to as millimeter wave (mmWave) including 28 GHz and 39 GHz. In addition, it has been considered to implement sixth generation (6G) mobile communication technologies (referred to as Beyond 5G systems) in terahertz (THz) bands (for example, 95 GHz to 3 THz bands) in order to accomplish transmission rates fifty times faster than 5G mobile communication technologies and ultra-low latencies one-tenth of 5G mobile communication technologies.

(4) At the beginning of the development of 5G mobile communication technologies, in order to support services and to satisfy performance requirements in connection with enhanced Mobile BroadBand (eMBB), Ultra Reliable Low Latency Communications (URLLC), and massive Machine-Type Communications (mMTC), there has been ongoing standardization regarding beamforming and massive multiple-input multiple-output (MIMO) for mitigating radio-wave path loss and increasing radio-wave transmission distances in mmWave, supporting numerologies (for example, operating multiple subcarrier spacings) for efficiently utilizing mmWave resources and dynamic operation of slot formats, initial access technologies for supporting multi-beam transmission and broadbands, definition and operation of BandWidth Part (BWP), new channel coding methods such as a Low Density Parity Check (LDPC) code for large amount of data transmission and a polar code for highly reliable transmission of control information, L2 pre-processing, and network slicing for providing a dedicated network specialized to a specific service.

(5) Currently, there are ongoing discussions regarding improvement and performance enhancement of initial 5G mobile communication technologies in view of services to be supported by 5G mobile communication technologies, and there has been physical layer standardization regarding technologies such as Vehicle-to-everything (V2X) for aiding driving determination by autonomous vehicles based on information regarding positions and states of vehicles transmitted by the vehicles and for enhancing user convenience, New Radio Unlicensed (NR-U) aimed at system operations conforming to various regulation-related requirements in unlicensed bands, NR user equipment (UE) Power Saving, Non-Terrestrial Network (NTN) which is UE-satellite direct communication for providing coverage in an area in which communication with terrestrial networks is unavailable, and positioning.

(6) Moreover, there has been ongoing standardization in air interface architecture/protocol regarding technologies such as Industrial Internet of Things (IIoT) for supporting new services through interworking and convergence with other industries, Integrated Access and Backhaul (IAB) for providing a node for network service area expansion by supporting a wireless backhaul link and an access link in an integrated manner, mobility enhancement including conditional handover and Dual Active Protocol Stack (DAPS) handover, and two-step random access for simplifying random access procedures (2-step random access channel (RACH) for NR). There also has been ongoing standardization in system architecture/service regarding a 5G baseline architecture (for example, service based architecture or service based interface) for combining Network Functions Virtualization (NFV) and Software-Defined Networking (SDN) technologies, and Mobile Edge Computing (MEC) for receiving services based on UE positions.

(7) As 5G mobile communication systems are commercialized, connected devices that have been exponentially increasing will be connected to communication networks, and it is accordingly expected that enhanced functions and performances of 5G mobile communication systems and integrated operations of connected devices will be necessary. To this end, new research is scheduled in connection with eXtended Reality (XR) for efficiently supporting Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR) and the like, 5G performance improvement and

complexity reduction by utilizing Artificial Intelligence (AI) and Machine Learning (ML), AI service support, metaverse service support, and drone communication.

(8) Furthermore, such development of 5G mobile communication systems will serve as a basis for developing not only new waveforms for providing coverage in terahertz bands of 6G mobile communication technologies, multi-antenna transmission technologies such as Full Dimensional MIMO (FD-MIMO), array antennas and large-scale antennas, metamaterial-based lenses and antennas for improving coverage of terahertz band signals, high-dimensional space multiplexing technology using Orbital Angular Momentum (OAM), and Reconfigurable Intelligent Surface (RIS), but also full-duplex technology for increasing frequency efficiency of 6G mobile communication technologies and improving system networks, AI-based communication technology for implementing system optimization by utilizing satellites and Artificial Intelligence (AI) from the design stage and internalizing end-to-end AI support functions, and next-generation distributed computing technology for implementing services at levels of complexity exceeding the limit of UE operation capability by utilizing ultra-high-performance communication and computing resources.

(9) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

(10) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a dual polarization patch antenna structure having a high isolation characteristic and an electronic device including same.

(11) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(12) In accordance with an aspect of the disclosure, an antenna module in a wireless communication system is provided. The antenna module includes a multi-layer printed circuit board (PCB) including a first to a sixth layer, a first feed line disposed in a first area of one side surface of the fifth layer of the multi-layer PCB, a second feed line disposed in a second area of the one side surface of the fifth layer to face the first feed line, a first via extending through the first area of the fifth layer and a first area of the fourth layer of the multi-layer PCB, and having one end electrically connected to the first feed line and the other end extending by a first height (h_1) in a vertical direction from the first area of the fourth layer, a second via extending through the second area of the fifth layer and a second area of the fourth layer while facing the first via, and having one end electrically connected to the second feed line and the other end extending by the first height (h_1) in a vertical direction from the second area of the fourth layer, a third feed line having one end electrically connected to the other end of the first via, and the other end extending horizontally in a first direction from the other end of the first via, a fourth feed line facing the third feed line, having one end electrically connected to the other end of the second via, and extending horizontally in a second direction from the other end of the second via, a third via having one end electrically connected to the other end of the third feed line, and the other end configured to extend through a fourth area of the third layer and extend by a second height (h_2) in a vertical direction, a fourth via facing the third via, and having one end electrically connected to the other end of the fourth feed line and the other end configured to extend through the fourth area of the fourth layer and extend by the second height (h_2) in a vertical direction, a first antenna patch having a lower surface electrically connected to the other end of the third via and the other end of the fourth via, and a second antenna patch spaced apart by a predetermined third height (h_3) from an upper surface of the first antenna patch. A spaced distance between the first via and the second via may be greater than a spaced distance between the third via and the fourth via.

(13) The first via includes a (1a)th via pad disposed in the first area of the fifth layer and physically coupled to one end of the first via, a (1b)th via pad disposed in the first area of the fourth layer and

physically coupled to a middle step of the first via, and a (1c)th via pad physically connected to the other end of the first via.

(14) The second via includes a (2a)th via pad disposed in the second area of the fifth layer and physically coupled to one end of the second via, a (3b)th via pad disposed in the second area of the fifth layer and physically coupled to a middle step of the second via, and a (3c)th via pad physically connected to the other end of the second via.

(15) The third via includes a (3a)th via pad physically coupled to one end of the third via, and a (3b)th via pad physically coupled to the other end of the third via and embedded in a lower surface of a probe pad.

(16) The fourth via includes a (4a)th via pad physically coupled to one end of the fourth via, and a (4b)th via pad physically coupled to the other end of the fourth via and embedded in a second area of a lower surface of a probe pad.

(17) The first feed line and the second feed line may be electrically connected to a radio frequency integrated circuit (RFIC).

(18) A first signal output from the RFIC may be emitted through the second antenna patch via the first feed line, the first via, the third feed line, the third via, and a first area of the first antenna patch.

(19) A second signal output from the RFIC may be emitted through the second antenna patch via the second feed line, the second via, the fourth feed line, the fourth via, and a second area of the first antenna patch.

(20) A first shielding structure configured to surround a first side surface, a second side surface, and an upper surface of the first via, and a second shielding structure configured to surround a first side surface, a second side surface, and an upper surface of the third via may be included therein. The first shielding structure and the second shielding structure may be arranged symmetrically with each other.

(21) The first shielding structure includes a first shielding via spaced apart by a predetermined distance from the first side surface of the first via, a second shielding via spaced apart by a predetermined distance from the second side surface of the first via, and a first shielding cover configured to extend from one end of the first shielding via to one end of the second shielding via and spaced apart by a predetermined distance from the upper surface of the first via.

(22) The first shielding via may be configured to extend from the third layer to a height which is substantially equal to the first antenna patch. The second shielding via may be configured to extend from the third layer to a height which is substantially equal to the first antenna patch.

(23) The first shielding cover may be disposed at a height which is substantially equal to the first antenna patch so as to be substantially parallel to a first side surface of the first antenna patch.

(24) The second shielding structure includes a third shielding via spaced apart by a predetermined distance from the first side surface of the second via, a fourth shielding via spaced apart by a predetermined distance from the second side surface of the third via, and a second shielding cover configured to extend from one end of the third shielding via to one end of the fourth shielding via and spaced apart by a predetermined distance from the upper surface of the second via.

(25) The third shielding via may be configured to extend from the third layer to a height which is substantially equal to the first antenna patch. The fourth shielding via may be configured to extend from the third layer to a height which is substantially equal to the first antenna patch.

(26) The second shielding cover may be disposed at a height which is substantially equal to the first antenna patch so as to be substantially parallel to the second side surface of the first antenna patch.

(27) A third shielding structure configured to surround a third side surface of the first via, and a fourth shielding structure configured to surround a third side surface of the third via may be included therein. The third shielding structure and the fourth shielding structure may be arranged symmetrically with each other.

(28) The third shielding structure includes a fifth shielding via spaced apart by a predetermined

distance from the third side surface of the first via, and a third shielding cover configured to extend from one end of the fifth shielding via to the first shielding structure.

(29) The fifth shielding via may be configured to extend from the third layer to a height which is substantially equal to the first antenna patch. The third shielding cover may be disposed at a height which is substantially equal to the first antenna patch so as to be substantially perpendicular to a first side surface of the first antenna patch on a x-y plane.

(30) The fourth shielding structure includes a sixth shielding via spaced apart by a predetermined distance from the third side surface of the third via, and a fourth shielding cover configured to extend from one end of the sixth shielding via to the second shielding structure.

(31) The sixth shielding via may be configured to extend from the third layer to a height which is substantially equal to a probe pad.

(32) The fourth shielding cover may be disposed at a height which is substantially equal to the probe pad so as to be substantially perpendicular to a second side surface of the probe pad on an x-y plane.

(33) According to the disclosure, isolation characteristics between antenna ports can be improved through a dual polarization patch antenna structure and an electronic device including same.

(34) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view showing an antenna structure according to an embodiment of the disclosure;
- (3) FIG. 2 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;
- (4) FIG. 3 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure;
- (5) FIG. 4 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;
- (6) FIG. 5 is a perspective view showing an antenna structure according to an embodiment of the disclosure;
- (7) FIG. 6 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;
- (8) FIG. 7 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure;
- (9) FIG. 8 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;
- (10) FIG. 9 is a perspective view showing an antenna structure according to an embodiment of the disclosure;
- (11) FIG. 10 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;
- (12) FIG. 11 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;
- (13) FIG. 12 is a cross-sectional view showing an antenna structure with reference to the x-z axis

according to an embodiment of the disclosure;

(14) FIG. 13 is a perspective view showing an antenna structure including a first shielding structure and a second shielding structure according to an embodiment of the disclosure;

(15) FIG. 14 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;

(16) FIG. 15 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;

(17) FIG. 16 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure;

(18) FIG. 17 is a perspective view showing an antenna structure including a third shielding structure and a fourth shielding structure according to an embodiment of the disclosure;

(19) FIG. 18 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;

(20) FIG. 19 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;

(21) FIG. 20 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure;

(22) FIG. 21 is a perspective view showing an antenna structure including a fifth shielding structure and a sixth shielding structure according to an embodiment of the disclosure;

(23) FIG. 22 is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure;

(24) FIG. 23 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure;

(25) FIG. 24 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure; and

(26) FIG. 25 is a graph showing an isolation characteristic between antenna ports and of the antenna structure according to an embodiment of the disclosure.

(27) Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

(28) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

(29) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(30) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

(31) FIG. 1 is a perspective view showing an antenna structure according to an embodiment of the disclosure.

(32) Referring to FIG. 1, an antenna structure 10 may include a multi-layer printed circuit board (PCB) 100, a first feed line 210, a second feed line 220, and a first via 310, a second via 320, a first

antenna patch **400**, and a second antenna patch **500**. The first antenna patch **400** may be referred to as a lower antenna patch. The second antenna patch **500** may be referred to as an upper antenna patch.

(33) The multi-layer PCB **100** may include a first substrate **111**, a second substrate **121**, a third substrate **131**, a first dielectric layer **112**, a second dielectric layer **122**, and a third dielectric layer **132**. For example, the first dielectric layer **112** may be disposed on the upper surface of the first substrate **111**. The second substrate **121** may be disposed on the upper surface of the first dielectric layer **112**. The second dielectric layer **122** may be disposed on the upper surface of the second substrate **121**. The third substrate **131** may be disposed on the upper surface of the second dielectric layer **122**. The third dielectric layer **132** may be disposed on the upper surface of the third substrate **131**.

(34) The first feed line **210** may be disposed in a (2a)th groove **121a** of the second substrate **121**. The (2a)th groove **121a** may be referred to as a (2a)th area **121a**. The first feed line **210** or one end of the first feed line **210** may be referred to as a first port. One end of the first feed line **210** may be electrically connected to a separate first port (not shown).

(35) The other end of the first feed line **210** may be electrically connected to one end of the first via **310**. The other end of the first feed line **210** may be electrically connected to one end of the first via **310** in a (2c)th groove **121c** of the second substrate **121**. The (2c)th groove **121c** may be referred to as a (2c)th area **121c**.

(36) The second feed line **220** may be disposed in a (2b)th groove **121b** of the second substrate **121**. The (2b)th groove **121b** may be referred to as a (2b)th area **121b**. The second feed line **220** or one end of the second feed line **220** may be referred to as a second port. One end of the second feed line **220** may be electrically connected to a separate second port (not shown).

(37) The other end of the second feed line **220** may be electrically connected to one end of the second via **320**. The other end of the second feed line **220** may be electrically connected to the other end of the second via **320** in a (2d)th groove **121d** of the second substrate **121**. The (2d)th groove **121d** may be referred to as a (2d)th area **121d**.

(38) The first via **310** may be disposed at the lower end of the first antenna patch **400**. The other end of the first via **310** may be electrically connected to a first area **401** of the lower surface of the first antenna patch **400**. The first via **310** may extend through a part of each of the second dielectric layer **122**, the third substrate **131**, and the third dielectric layer **132** to be electrically connected to the first area **401** of the lower surface of the first antenna patch **400**. For example, the first via **310** may extend through a (3c)th groove **131c** of the third substrate **131** and a part of the third dielectric layer **132** to be electrically connected to the first area **301** of the lower surface of the first antenna patch **400**. The first via **310** may be referred to as a first probe feed.

(39) The second via **320** may be disposed at the lower end of the second antenna patch **500**. The other end of the second via **320** may be electrically connected to a second area **402** of the lower surface of the second antenna patch **500**. The second via **320** may extend through a part of each of the second dielectric layer **122**, the third substrate **131**, and the third dielectric layer **132** to be electrically connected to the second area **402** of the lower surface of the first antenna patch **400**. For example, the second via **320** may extend through a (3d)th groove **131d** of the third substrate **131** and a part of the third dielectric layer **132** to be electrically connected to the second area **402** of the lower surface of the first antenna patch **400**. The second via may be referred to as a second probe feed.

(40) The first antenna patch **400** may be disposed on the lower end of the second antenna patch **500**. The first antenna patch **400** may be disposed between the third substrate **131** and the second antenna patch **500**. The first antenna patch **400** may be disposed inside the third dielectric layer **132**. The first antenna patch **400** may be physically separated from the second antenna patch **500**. The first antenna patch **400** may be electromagnetically coupled to the second antenna patch **500**.

(41) The second antenna patch **500** may be disposed on the upper surface of the multi-layer PCB

100. The second antenna patch **500** may be disposed on the upper surface of the third dielectric layer **132**.

(42) FIG. **2** is a plan view showing an antenna structure **10** with reference to the x-y axis according to an embodiment of the disclosure.

(43) Referring to FIG. **2**, a horizontal length w_1 of the multi-layer PCB **100** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that the operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_1 of the multi-layer PCB **100** may be approximately 910 μm . A vertical length d_1 of the multi-layer PCB **100** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the vertical length d_1 of the multi-layer PCB **100** may be approximately 910 μm .

(44) The first antenna patch **400** and the second antenna patch **500** may be arranged to overlap each other in the z-axis direction. A horizontal length w_5 of the second antenna patch **500** may be equal to or different from a horizontal length w_4 of the first antenna patch **400**. For example, the horizontal length w_5 of the second antenna patch **500** may be greater than the horizontal length w_4 of the first antenna patch **400**, but it is not limited thereto. For example, the horizontal length w_5 of the second antenna patch **500** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_5 of the second antenna patch **500** may be approximately 490 μm . For example, the horizontal length w_4 of the first antenna patch **400** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_4 of the first antenna patch **400** may be approximately 460 μm . The vertical length of the second antenna patch **500** may be equal to or different from the vertical length of the first antenna patch **400**. For example, the vertical length of the second antenna patch **500** may be greater than the vertical length of the first antenna patch **400**, but it is not limited thereto.

(45) The (2c)th groove **121c** of the second substrate **121** and the (3c)th groove **131c** of the third substrate **131** may be arranged to overlap each other in the z-axis direction. A diameter w_3 of the (2c)th groove **121c** of the second substrate **121** may be equal to or similar to the diameter of the (3c)th groove **131c** of the third substrate **131**. For example, the diameter w_3 of the (2c)th groove **121c** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter w_3 of the (2c)th groove **121c** may be approximately 120 μm .

(46) The (2d)th groove **121d** of the second substrate **121** and the (3d)th groove **131d** of the third substrate **131** may be arranged to overlap each other in the z-axis direction. A diameter w_3 of the (2d)th groove **121d** of the second substrate **121** may be equal to or similar to the diameter of the (3d)th groove **131d** of the third substrate **131**. For example, the diameter w_3 of the (2d)th groove **121d** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof.

(47) A diameter w_2 of the cross section of the first via **310** with reference to the x-y axis may be smaller than the diameter w_3 of the (2c)th groove **121c** of the second substrate **121** or the diameter of the (3c)th groove **131c** of the third substrate **131**. For example, the diameter w_2 of the cross section of the first via **310** with reference to the x-y axis may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna

structure **10** is 140 GHz, the diameter w2 of the cross section of the first via **310** with reference to the x-y axis may be approximately 40 μm .

(48) The diameter w2 of the cross section of the second via **320** with reference to the x-y axis may be smaller than the diameter w3 of the (2d)th groove **121d** of the second substrate **121** or the diameter of the (3d)th groove **131d** of the third substrate **131**. For example, the diameter w2 of the cross section of the second via **320** with reference to the x-y axis may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof.

(49) The width w2 of the first feed line **210** may be smaller than the width w3 of the (2a)th groove **121a**. The width of the second feed line **220** may be smaller than the width of the (2b)th groove **121b**. For example, the width w2 of the first feed line **210** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, the width w2 of the second feed line **220** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof.

(50) FIG. **3** is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure.

(51) FIG. **4** is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(52) Referring to FIGS. **3** and **4**, the multi-layer PCB **100** may include a first to a fifth layer. For example, the first layer of the multi-layer PCB **100** may correspond to the second antenna patch **500**. The second layer of the multi-layer PCB **100** may correspond to the first antenna patch **400**. The third layer of the multi-layer PCB **100** may correspond to the third substrate **131**. The fourth layer of the multi-layer PCB **100** may correspond to the second substrate **121**. The fifth layer of the multi-layer PCB **100** may correspond to the first substrate **111**.

(53) The first substrate **111** and the second substrate **121** may be spaced apart by a predetermined height h1 from each other. For example, the predetermined height h1 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h1 may be approximately 50 μm . The first dielectric layer **112** may be filled between the first substrate **111** and the second substrate **121**.

(54) The second substrate **121** and the third substrate **131** may be spaced apart by a predetermined height h2 from each other. For example, the predetermined height h2 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h2 may be approximately 50 μm . The second dielectric layer **122** may be filled between the second substrate **121** and the third substrate **131**.

(55) The third dielectric layer **132** may be filled on the upper surface of the third substrate **131**. The first antenna patch **400** may be embedded in the third dielectric layer **132**. The first antenna patch **400** may be disposed to be spaced apart by a predetermined height h4 from the third substrate **131**. For example, the predetermined height h4 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h4 may be approximately 115 μm .

(56) The second antenna patch **500** may be disposed to be spaced apart by a predetermined height h5 from the first antenna patch **400**. For example, the predetermined height h5 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an

operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h_5 may be approximately 50 μm .

(57) The thickness of each of the first antenna patch **400**, the second antenna patch **500**, the first substrate **111**, the second substrate **121**, the third substrate **131**, the first dielectric layer **112**, and the second dielectric layer **122**, that is, the height thereof with reference to the z-axis may be approximately 15 μm .

(58) The first via **310** may electrically connect the first antenna patch **400** and the first feed line **210**. One end of the first feed line **210** may be electrically connected to the first port. The other end of the first feed line **210** may be electrically connected to one end of the first via **310**.

(59) The second via **320** may electrically connect the first antenna patch **400** and the second feed line **220**. One end of the second feed line **220** may be electrically connected to the second port. The other end of the second feed line **220** may be electrically connected to one end of the second via **320**.

(60) The (2a)th groove **121a** and the (2b)th groove **121b** of the second substrate **121** may be spaced apart by a predetermined distance w_6 from each other in the x-axis direction. For example, the predetermined distance w_6 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_6 may be approximately 70 μm .

(61) The (3c)th groove **131c** and the (3d)th groove **131d** of the third substrate **131** may be spaced apart by the predetermined distance w_6 from each other in the x-axis direction. For example, the predetermined distance w_6 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof.

(62) The first via **310** and the second via **320** may be spaced apart by a predetermined distance w_7 from each other in the x-axis direction. For example, the predetermined distance w_7 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_7 may be approximately 130 μm .

(63) FIG. 5 is a perspective view showing an antenna structure according to an embodiment of the disclosure.

(64) Referring to FIG. 5, the antenna structure **10** may include a multi-layer PCB **100**, a first feed line **210**, a second feed line **220**, a first via **310**, a second via **320**, a (1a)th via pad **310a**, a (1b)th via pad **310b**, a (1c)th via pad **310c**, a (1d)th via pad **310d**, a (2a)th via pad **320a**, a (2b)th via pad **320b**, a (2c)th via pad **320c**, a (2d)th via pad **320d**, a first antenna patch **400**, and a second antenna patch **500**.

(65) The first via **310** may be physically coupled to the (1a)th via pad **310a**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the (1d)th via pad **310d**. For example, the first via **310** may be divided into a first step to a fourth step with reference to the z-axis. For example, the (1a)th via pad **310a** may be physically coupled to the first step of the first via **310**. The (1b)th via pad **310b** may be physically coupled to the second step of the first via **310**. The (1c)th via pad **310c** may be physically coupled to the third step of the first via **310**. The (1d)th via pad **310d** may be physically coupled to the fourth step of the first via **310**. The first via **310** may be referred to as a first probe feed.

(66) The second via **320** may be physically coupled to the (2a)th via pad **320a**, the (2b)th via pad **320b**, the (2c)th via pad **320c**, and the (2d)th via pad **320d**. For example, the second via **320** may be divided into a first step to a fourth step with reference to the z-axis. For example, the (2a)th via pad **320a** may be physically coupled to the first step of the second via **320**. The (2b)th via pad **320b** may be physically coupled to the second step of the second via **320**. The (2c)th via pad **320c** may

be physically coupled to the third step of the second via **320**. The (2d)th via pad **320d** may be physically coupled to the fourth step of the second via **320**. The second via **320** may be referred to as a second probe feed.

(67) The (1a)th via pad **310a** may be disposed in the (2c)th groove **121c** of the second substrate **121**. The (1b)th via pad **310b** may be disposed in the (3c)th groove **131c** of the third substrate **131**. The (1d)th via pad **310d** may be physically coupled to the first antenna patch **400**. The first antenna patch **400** may include the (1d)th via pad **310d**.

(68) The (2a)th via pad **320a** may be disposed in the (2d)th groove **121d** of the second substrate **121**. The (2b)th via pad **320b** may be disposed in the (3d)th groove **131d** of the third substrate **131**. The (2d)th via pad **320d** may be physically coupled to the first antenna patch **400**. The first antenna patch **400** may include the (2d)th via pad **320d**.

(69) FIG. **6** is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure.

(70) Referring to FIG. **6**, the (1a)th via pad **310a**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the (1d)th via pad **310d** may be arranged to overlap each other in the z-axis direction. A diameter r_1 of the (1a)th via pad **310a**, a diameter of the (1b)th via pad **310b**, a diameter of the (1c)th via pad **310c**, and a diameter of the (1d)th via pad **310d** may be mutually equal or similar. For example, the diameter r_1 of the (1a)th via pad **310a** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r_1 of the (1a)th via pad **310a** may be approximately 130 μm .

(71) The (2a)th via pad **320a**, the (2b)th via pad **320b**, the (2c)th via pad **320c**, and the (2d)th via pad **320d** may be arranged to overlap each other in the z-axis direction. A diameter of the (2a)th via pad **320a**, a diameter of the (2b)th via pad **320b**, a diameter of the (2c)th via pad **320c**, and a diameter of the (2d)th via pad **320d** may be mutually equal or similar. For example, the diameter of the (2a)th via pad **320a** may be equal to or similar to the diameter r_1 of the (1a)th via pad **310a**.

(72) The (2c)th groove **121c** of the second substrate **121** and the (3c)th groove **131c** of the third substrate **131** may be arranged to overlap each other in the z-axis direction. A diameter r_2 of the (2c)th groove **121c** of the second substrate **121** may be equal to or similar to the diameter of the (3c)th groove **131c** of the third substrate **131**. For example, the diameter r_2 of the (2c)th groove **121c** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r_2 of the (2c)th groove **121c** may be approximately 210 μm .

(73) The (2d)th groove **121d** of the second substrate **121** and the (3d)th groove **131d** of the third substrate **131** may be arranged to overlap each other in the z-axis direction. The diameter of the (2d)th groove **121d** of the second substrate **121** may be equal to or similar to the diameter of the (3d)th groove **131d** of the third substrate **131**. For example, the diameter of the (2d)th groove **121d** may be equal to or similar to the diameter r_2 of the (2c)th groove **121c**.

(74) The diameter r_1 of each of the (1a)th via pad **310a**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the (1d)th via pad **310d** may be smaller than the diameter r_2 of the (2c)th groove **121c** of the second substrate **121** or the diameter of the (3c)th groove **131c** of the third substrate **131**.

(75) The diameter of each of the (2a)th via pad **320a**, the (2b)th via pad **320b**, the (2c)th via pad **320c**, and the (2d)th via pad **320d** may be smaller than the diameter of the (2b)th groove **121b** of the second substrate **121** or the diameter of the (3b)th groove **131b** of the third substrate **131**.

(76) The antenna structure **10** according to the disclosure may further include a ground via. For example, the ground via may be disposed between the first feed line **210** and the second feed line **220**. The ground via may be disposed to overlap a partial area of the (2a)th groove **121a** of the second substrate **121** and a partial area of the (2b)th groove **121b** of the second substrate **121**, in the z-axis direction.

(77) The width $r3$ of the ground via may be equal to or similar to the diameter $r1$ of each of the (1a)th via pad **310a**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the (1d)th via pad **310d**, or the diameter of each of the (2a)th via pad **320a**, the (2b)th via pad **320b**, the (2c)th via pad **320c**, and the (2d)th via pad **320d**.

(78) FIG. 7 is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure.

(79) FIG. 8 is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(80) Referring to FIGS. 7 and 8, the multi-layer PCB **100** may include a first layer to a sixth layer. For example, the first layer of the multi-layer PCB **100** may correspond to the second antenna patch **500**. The second layer of the multi-layer PCB **100** may correspond to the first antenna patch **400**. The third layer of the multi-layer PCB **100** may correspond to at least one of the (1c)th via pad **310c** and the (2c)th via pad **320c**. The fourth layer of the multi-layer PCB **100** may correspond to the third substrate **131**. The fifth layer of the multi-layer PCB **100** may correspond to the second substrate **121**. The sixth layer of the multi-layer PCB **100** may correspond to the first substrate **111**.

(81) The first via **310** may extend from the second substrate **121** to the first antenna patch **400**. The (1a)th via pad **310a** may be disposed in the (2a)th groove **121a** of the second substrate **121**. The (1b)th via pad **310b** may be disposed in (3a)th groove **131a** of third substrate **131**. The (1b)th via pad **310b** may be spaced apart by the predetermined height $h2$ in the z-axis direction from the (1a)th via pad **310a**. For example, the predetermined height $h2$ may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height $h2$ may be approximately 50 μm . The (1c)th via pad **310c** may be arranged to be spaced apart by the predetermined height $h3$ in the z-axis direction from the (1b)th via pad **310b**. For example, the predetermined height $h3$ may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height $h3$ may be approximately 50 μm . The (1d)th via pad **310d** may be arranged to be spaced apart by the predetermined height $h4$ in the z-axis direction from the (1c)th via pad **310c**. For example, the predetermined height $h4$ may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height $h4$ may be approximately 50 μm .

(82) The second via **320** may extend from the second substrate **121** to the first antenna patch **400**. The (2a)th via pad **320a** may be disposed in the (2b)th groove **121b** of the second substrate **121**. The (2b)th via pad **320b** may be disposed in the (3b)th groove **131b** of the third substrate **131**. The (2b)th via pad **320b** may be spaced apart by the predetermined height $h2$ in the z-axis direction from the (2a)th via pad **320a**. The (2c)th via pad **320c** may be arranged to be spaced apart by the predetermined height $h3$ in the z-axis direction from the (2b)th via pad **320b**. The (2d)th via pad **320d** may be arranged to be spaced apart by the predetermined height $h4$ in the z-axis direction from the (2c)th via pad **320c**.

(83) The ground via includes a first ground via **610**, a second ground via **620**, a (1a)th ground pad **610a**, a (1b)th ground pad **610b**, a (1c)th ground pad **610c**, a (2a)th ground pad **620a**, a (2b)th ground pad **620b**, and a (2c)th ground pad **620c**.

(84) The first ground via **610** may extend through the first substrate **111**, the second substrate **121**, and the third substrate **131**. For example, one end of the first ground via **610** may be physically connected to the first substrate **111**. The other end of the first ground via **610** may be physically connected to the third substrate **131**.

(85) The (1a)th ground pad **610a** may be disposed on the first step of the first ground via **610**. The

first step of the first ground via **610** may correspond to the first substrate **111**. The (1a)th ground pad **610a** may be disposed on the first substrate **111**.

(86) The (1b)th ground pad **610b** may be disposed on the second step of the first ground via **610**. The second step of the first ground via **610** may correspond to the second substrate **121**. The (1b)th ground pad **610b** may be disposed on the second substrate **121**.

(87) The (1c)th ground pad **610c** may be disposed on the third step of the first ground via **610**. The third step of the first ground via **610** may correspond to the third substrate **131**. The (1c)th ground pad **610c** may be disposed on the third substrate **131**.

(88) The second ground via **620** may extend through the first substrate **111**, the second substrate **121**, and the third substrate **131**. For example, one end of the second ground via **620** may be physically connected to the first substrate **111**. The other end of the second ground via **620** may be physically connected to the third substrate **131**.

(89) The (2a)th ground pad **620a** may be disposed on the first step of the second ground via **620**. The first step of the second ground via **620** may correspond to the first substrate **111**. The (2a)th ground pad **620a** may be disposed on the first substrate **111**.

(90) The (2b)th ground pad **620b** may be disposed on the second step of the second ground via **620**. The second step of the second ground via **620** may correspond to the second substrate **121**. The (2b)th ground pad **620b** may be disposed on the second substrate **121**.

(91) The (2c)th ground pad **620c** may be disposed on the third step of the second ground via **620**. The third step of the second ground via **620** may correspond to the third substrate **131**. The (2c)th ground pad **620c** may be disposed on the third substrate **131**.

(92) The thickness of each of the first antenna patch **400**, the second antenna patch **500**, the first substrate **111**, the second substrate **121**, the third substrate **131**, the first dielectric layer **112**, the second dielectric layer **122**, the (1c)th via pad **310c**, and the (1d)th via pad **310d**, that is, the height thereof with reference to the z-axis may be approximately 15 μm .

(93) The (1a)th via pad **310a** and the (2a)th via pad **320a** may be spaced apart by a predetermined distance w_{12} from each other in the x-axis direction. For example, the predetermined distance w_{12} may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_{12} may be approximately 70 μm .

(94) The first via **310** and the second via **320** may be spaced apart by a predetermined distance w_{13} from each other in the x-axis direction. For example, the predetermined distance w_{13} may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_{13} may be approximately 130 μm .

(95) FIG. 9 is a perspective view showing an antenna structure according to an embodiment of the disclosure.

(96) Referring to FIG. 9, the antenna structure **10** may include a multi-layer PCB **100**, a first feed line **210**, a second feed line **220**, a third feed line **230**, a fourth feed line **240**, a first impedance matching structure **250**, a second impedance matching structure **260**, a first via **310**, a second via **320**, a third via **330**, a fourth via **340**, a (1a)th via pad **310a**, a (1b)th via pad **310b**, a (1c)th via pad **310c**, a (2a)th via pad **320a**, a (2b)th via pad **320b**, a (3a)th via pad **330a**, a (3b)th via pad **330b**, a (3c)th via pad **330c**, a (4a)th via pad **340a**, a (4b)th via pad **340b**, a first antenna patch **400**, and a second antenna patch **500**.

(97) The first feed line **210** may be disposed in the (2a)th groove **121a** of the second substrate **121**. One end of the first feed line **210** may be referred to as a first port. One end of the first feed line **210** may be electrically connected to a separate first port (not shown).

(98) The other end of the first feed line **210** may be electrically connected to the (1a)th via pad

310a. The other end of the first feed line **210** may be electrically connected to one end of the first via **310** in the (2c)th groove **121c** of the second substrate **121**.

(99) The first via **310** may be physically coupled to the (1a)th via pad **310a**, the (1b)th via pad **310b**, and the (1c)th via pad **310c**. For example, the first via **310** may be divided into a first step to a third step with reference to the z-axis. For example, the (1a)th via pad **310a** may be physically coupled to the first step of the first via **310**. The (1b)th via pad **310b** may be physically coupled to the second step of the first via **310**. The (1c)th via pad **310c** may be physically coupled to the third step of the first via **310**.

(100) The (1a)th via pad **310a** may be disposed in the (2c)th groove **121c** of the second substrate **121**. The (1b)th via pad **310b** may be disposed in the (3c)th groove **131c** of the third substrate **131**. The (1c)th via pad **310c** may be electrically connected to one end of the third feed line **230**.

(101) The first impedance matching structure **250** may have a line shape extending from each of one side and the other side of the (1a)th via pad **310a** in the (2c)th groove **121c** of the second substrate **121**.

(102) One end of the second feed line **220** may be electrically connected to the (1c)th via pad **310c**. The other end of the second feed line **220** may be electrically connected to the (2a)th via pad **320a**.

(103) The second via **320** may include the (2a)th via pad **320a** and the (2b)th via pad **320b**. For example, the second via **320** may be divided into a first step and a second step with reference to the z-axis. For example, the (2a)th via pad **320a** may be physically coupled to the first step of the second via **320**. The (2b)th via pad **320b** may be physically coupled to the second step of the second via **320**. The (2b)th via pad **320b** may be physically coupled to the first antenna patch **400**. The first antenna patch **400** may include the (2b)th via pad **320b**.

(104) The second feed line **220** may be disposed in a (2b)th groove **121b** of the second substrate **121**. One end of the second feed line **220** may be referred to as a second port. One end of the second feed line **220** may be electrically connected to a separate second port (not shown).

(105) The other end of the third feed line **230** may be electrically connected to the (3a)th via pad **330a**. The other end of the third feed line **230** may be electrically connected to one end of the third via **330** in the (2c)th groove **121c** of the second substrate **121**.

(106) The third via **330** may be physically coupled to the (3a)th via pad **330a**, the (3b)th via pad **330b**, and the (3c)th via pad **330c**. For example, the third via **330** may be divided into a first step to a third step with reference to the z-axis. For example, the (3a)th via pad **330a** may be physically coupled to the first step of the third via **330**. The (3b)th via pad **330b** may be physically coupled to the second step of the third via **330**. The (3c)th via pad **330c** may be physically coupled to the third step of the third via **330**.

(107) One end of the fourth feed line **240** may be electrically connected to the (3c)th via pad **330c**. The other end of the fourth feed line **240** may be electrically connected to the (4a)th via pad **340a**.

(108) The fourth via **340** may include the (4a)th via pad **340a** and the (4b)th via pad **340b**. For example, the fourth via **340** may be divided into a first step and a second step with reference to the z-axis. For example, the (4a)th via pad **340a** may be physically coupled to the first step of the fourth via **340**. The (4b)th via pad **340b** may be physically coupled to the second step of the fourth via **340**. The (4b)th via pad **340b** may be physically coupled to the first antenna patch **400**. The first antenna patch **400** may include the (4b)th via pad **340b**. The fourth via **340** may be referred to as a second probe feed.

(109) FIG. **10** is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure.

(110) Referring to FIG. **10**, the (1a)th via pad **310a**, the (1b)th via pad **310b**, and the (1c)th via pad **310c** may be arranged to overlap each other with reference to the z-axis. The (2a)th via pad **320a** and the (2b)th via pad **320b** may be arranged to overlap each other with reference to the z-axis.

(111) For example, the diameter r_4 of the first via **310**, the diameter r_5 of the (1c)th via pad **310c**, and the diameter r_6 of the (2a)th groove **121a** may be determined based on at least one of an

operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r_4 of the first via **310** may be approximately 40 μm to 70 μm . The diameter r_5 of the (1c)th via pad **310c** may be approximately 120 μm to 130 μm . The diameter (r_6) of the (2a)th groove **121a** may be approximately 210 μm .

(112) The third feed line **230** may extend from the (1c)th via pad **310c** toward the (2a)th via pad **320a**. For example, the horizontal length w_{14} of the third feed line **230** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_{14} of the third feed line **230** may be approximately 50 μm . The vertical length d_2 of the third feed line **230** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_{14} of the third feed line **230** may be approximately 235 μm . The horizontal length w_{14} of the third feed line **230** may be equal to or similar to the length from the center of the first via **310** to the center of the third via **330** on the x-y plane.

(113) The (3a)th via pad **330a**, the (3b)th via pad **330b**, and the (3c)th via pad **330c** may be arranged to overlap each other with reference to the z-axis. The (4a)th via pad **340a** and the (4b)th via pad **340b** may be arranged to overlap each other with reference to the z-axis.

(114) The fourth feed line **240** may extend from the (3c)th via pad **330c** toward the (4a)th via pad **340a**. For example, the horizontal length of the fourth feed line **240** may be equal to or similar to the horizontal length w_{14} of the third feed line **230**. The vertical length of the fourth feed line **240** may be equal to or similar to the vertical length d_2 of the third feed line **230**.

(115) The (1a)th via pad **310a** and the (3a)th via pad **330a** may be spaced apart by a predetermined distance from each other in the x-axis direction with reference to the x-y plane. For example, the predetermined distance may be equal to or similar to the vertical length d_2 of the third feed line **230**.

(116) The (1b)th via pad **310b** and the (3b)th via pad **330b** may be spaced apart by a predetermined distance from each other in the x-axis direction with reference to the x-y plane. For example, the predetermined distance may be equal to or similar to the vertical length of the fourth feed line **240**.

(117) The first impedance matching structure **250** may have a structure extending from one side and the other side of the (1a)th via pad **310a** with reference to the x-y plane. For example, the first impedance matching structure **250** may be disposed substantially perpendicular to the first feed line **210** with reference to the x-y plane. For example, the width w_{25} of the first impedance matching structure **250** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the width w_{25} of the first impedance matching structure **250** may be approximately 40 μm . The horizontal length of the first impedance matching structure **250** with reference to the x-y plane may have various values. For example, the impedance matching performance of the antenna structure **10** may vary according to the horizontal length of the first impedance matching structure **250** with reference the x-y plane.

(118) The second impedance matching structure **260** may have a structure extending from one side and the other side of the (2a)th via pad **320a** with reference to the x-y plane. For example, the second impedance matching structure **260** may be disposed substantially perpendicular to the second feed line **220** with reference to the x-y plane. For example, the width of the second impedance matching structure **260** may be equal to or similar to the width w_{24} of the first impedance matching structure **250**.

(119) FIG. **11** is a cross-sectional view showing an antenna structure with reference to the y-z axis

according to an embodiment of the disclosure.

(120) FIG. **12** is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(121) Referring to FIGS. **11** and **12**, the multi-layer PCB **100** may include a first layer to a sixth layer. For example, the first layer of the multi-layer PCB **100** may correspond to the second antenna patch **500**. The second layer of the multi-layer PCB **100** may correspond to the first antenna patch **400**. The third layer of the multi-layer PCB **100** may correspond to at least one of the (3a)th via pad **330a** and the (4a)th via pad **340a**. The fourth layer of the multi-layer PCB **100** may correspond to the third substrate **131**. The fifth layer of the multi-layer PCB **100** may correspond to the second substrate **121**. The sixth layer of the multi-layer PCB **100** may correspond to the first substrate **111**.

(122) The first via **310** may extend from the second substrate **121** to one end of the third feed line **230** in the z-axis direction. The (1b)th via pad **310b** may be spaced apart by the predetermined height h_2 in the z-axis direction from the (1a)th via pad **310a**. For example, the predetermined height h_2 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h_2 may be approximately 50 μm . The (1c)th via pad **310c** may be arranged to be spaced apart by the predetermined height h_3 in the z-axis direction from the (1b)th via pad **310b**. For example, the predetermined height h_3 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h_3 may be approximately 50 μm .

(123) The third feed line **230** may electrically connect the (1c)th via pad **310c** and the (2a)th via pad **320a**.

(124) The third via **330** may extend from the other end of the third feed line **230** to the first antenna patch **400**. The (3a)th via pad **330a** may be disposed at one end of the third via **330**. The (3b)th via pad **330b** may be disposed to be spaced apart by the predetermined height h_4 in the z-axis direction from the (3a)th via pad **330a**. For example, the predetermined height h_4 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined height h_4 may be approximately 50 μm .

(125) The second via **320** may extend from the second substrate **121** to one end of the fourth feed line **240** in the z-axis direction. The (3b)th via pad **330b** may be spaced apart by the predetermined height h_2 in the z-axis direction from the (3a)th via pad **330a**. The (3c)th via pad **330c** may be arranged to be spaced apart by the predetermined height h_3 in the z-axis direction from the (3b)th via pad **330b**.

(126) The fourth feed line **240** may electrically connect the (3c)th via pad **330c** and the (4a)th via pad **340a**.

(127) The second via **320** may extend from the other end of the fourth feed line **240** to the first antenna patch **400**. The (4a)th via pad **340a** may be disposed at one end of the fourth via **340**. The (4b)th via pad **340b** may be disposed to be spaced apart by the predetermined height h_4 in the z-axis direction from the (4a)th via pad **340a**.

(128) The first via **310** and the second via **320** may be spaced apart by a predetermined distance w_{15} from each other in the x-axis direction. For example, the predetermined distance w_{15} may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_{15} may be approximately 502 μm .

(129) The (1a)th via pad **310a** and the (2a)th via pad **320a** may be spaced apart by a predetermined

distance w16 from each other in the x-axis direction. For example, the predetermined distance w16 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w15 may be approximately 442 μm .

(130) The third via **330** and the fourth via **340** may be spaced apart by a predetermined distance w17 from each other in the x-axis direction. For example, the predetermined distance w17 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w15 may be approximately 130 μm .

(131) The (3a)th via pad **330a** and the (4a)th via pad **320a** may be spaced apart by a predetermined distance w18 from each other in the x-axis direction. For example, the predetermined distance w18 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w15 may be approximately 110 μm .

(132) FIG. **13** is a perspective view showing an antenna structure including a first shielding structure and a second shielding structure according to an embodiment of the disclosure.

(133) Referring to FIG. **13**, identically to the antenna structure **10** in FIGS. **9** to **12**, the antenna structure **10** may include a multi-layer PCB **100**, a first feed line **210**, a second feed line **220**, a third feed line **230**, a fourth feed line **240**, a first impedance matching structure **250**, a second impedance matching structure **260**, a first via **310**, a second via **320**, a third via **330**, a fourth via **340**, a (1a)th via pad **310a**, a (1b)th via pad **310b**, a (1c)th via pad **310c**, a (2a)th via pad **320a**, a (2b)th via pad **320b**, a (3a)th via pad **330a**, a (3b)th via pad **330b**, a (3c)th via pad **330c**, a (4a)th via pad **340a**, a (4b)th via pad **340b**, a first antenna patch **400**, and a second antenna patch **500**.

(134) The antenna structure **10** may include a first shielding structure and a second shielding structure. For example, the first shielding structure may be disposed to surround a part of each of the first via **310**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the second feed line **220**. The second shielding structure may be disposed to surround a part of each of the third via **330**, the (3b)th via pad **330b**, the (3c)th via pad **330c**, and the fourth feed line **240**.

(135) The first shielding structure may include a first shielding via **710**, a second shielding via **720**, a (1a)th shielding pad **710a**, a (1b)th shielding pad **710b**, a (1c)th shielding pad **710c**, a (2a)th shielding pad **720a**, a (2b)th shielding pad **720b**, a (2c)th shielding pad **720c**, and a first shielding cover **771**. The second shielding structure may include a third shielding via **730**, a fourth shielding via **740**, a (3a)th shielding pad **730a**, a (3b)th shielding pad **730b**, a (3c)th shielding pad **730c**, a (4a)th shielding pad **740a**, a (4b)th shielding pad **740b**, a (4c)th shielding pad **740c**, and a second shielding cover **772**.

(136) FIG. **14** is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure.

(137) Referring to FIG. **14**, the first shielding structure and the second shielding structure may be arranged to face each other on the x-y plane.

(138) The first shielding structure may be disposed substantially parallel to a first side surface of the first antenna patch **400**. The first shielding cover **771** may be disposed substantially parallel to the first side surface of the first antenna patch **400**. The first shielding structure may be spaced apart by a predetermined distance w19 from the first side surface of the first antenna patch **400**. The first shielding cover **771** may be spaced apart by the predetermined distance w19 from the first side surface of the first antenna patch **400**. For example, the predetermined distance w19 may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that

an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w19** may be approximately 105 μm .

(139) The center of the first shielding cover **771** may substantially coincide with the center of the first via **310** on the z-axis. One end of the first shielding cover **771** may be physically coupled to the (1c)th shielding pad **710c**. The other end of the first shielding cover **771** may be physically coupled to the (2c)th shielding pad **720c**. A horizontal length **w20** of the first shielding cover **771** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w20** may be approximately 470 μm . A vertical length **d3** of the first shielding cover **771** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the vertical length **d3** of the first shielding cover **771** may be approximately 130 μm .

(140) The (1c)th shielding pad **710c** may be spaced apart by a predetermined distance **w21** from (1c)th via pad **310c** on the x-y plane. For example, the predetermined distance **w21** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w21** may be approximately 40 μm . The (1a)th shielding pad **710a**, the (1b)th shielding pad **710b**, and the (1c)th shielding pad **710c** may be arranged to overlap each other on the z-axis.

(141) The (2c)th shielding pad **720c** may be spaced apart by a predetermined distance **w19** from (1c)th via pad **310c** on the x-y plane. The (2c)th shielding pad **720c** may be disposed to face the (1c)th via pad **710c** with reference to the (1c)th via pad **310c**. The (2a)th shielding pad **720a**, the (2b)th shielding pad **720b**, and the (2c)th shielding pad **720c** may be arranged to overlap each other on the z-axis.

(142) The second shielding structure may be disposed substantially parallel to a second side surface of the first antenna patch **400**. The second shielding cover **772** may be disposed substantially parallel to the second side surface of the first antenna patch **400**. The second shielding structure may be spaced apart by a predetermined distance from the second side surface of the first antenna patch **400**. For example, the spaced distance between the second shielding structure and the second side surface of the first antenna patch **400** may be equal to as or similar to the spaced distance **w19** between the first shielding structure and the first side surface of the first antenna patch **400**.

(143) The center of the second shielding cover **772** may substantially coincide with the center of the third via **330** on the z-axis. One end of the second shielding cover **772** may be physically coupled to the (3c)th shielding pad **730c**. The other end of the second shielding cover **772** may be physically coupled to the (3c)th shielding pad **730c**. For example, the horizontal length of the second shielding cover **772** may be equal to as or similar to the horizontal length **w20** of the first shielding cover **771**. The vertical length of the second shielding cover **772** may be equal to as or similar to the vertical length **d3** of the first shielding cover **771**.

(144) The (3c)th shielding pad **730c** may be spaced apart by a predetermined distance **w21** from (3c)th via pad **330c** on the x-y plane. The (3a)th shielding pad **730a**, the (3b)th shielding pad **730b**, and the (3c)th shielding pad **730c** may be arranged to overlap each other on the z-axis.

(145) The (4c)th shielding pad **740c** may be spaced apart by a predetermined distance **w21** from (3c)th via pad **330c** on the x-y plane. The (4c)th shielding pad **740c** may be disposed to face the (3c)th via pad **730c** with reference to the (3c)th via pad **330c**. The (4a)th shielding pad **740a**, the (4b)th shielding pad **740b**, and the (4c)th shielding pad **740c** may be arranged to overlap each other on the z-axis.

(146) FIG. **15** is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure.

(147) FIG. 16 is a cross-sectional view disclosing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(148) Referring to FIGS. 15 and 16, the multi-layer PCB 100 may include a first layer to a sixth layer. For example, the first layer of the multi-layer PCB 100 may correspond to the second antenna patch 500. The second layer of the multi-layer PCB 100 may correspond to at least one of the first antenna patch 400, the first shielding cover 771, and the second shielding cover 772. The third layer of the multi-layer PCB 100 may correspond to at least one of a (3a)th via pad 330a, a (4a)th via pad 340a, a (1b)th shielding pad 710b, a (2b)th shielding pad 720b, a (3b)th shielding pad 730b, and a (4b)th shielding pad 740b. The fourth layer of the multi-layer PCB 100 may correspond to the third substrate 131. The fifth layer of the multi-layer PCB 100 may correspond to the second substrate 121. The sixth layer of the multi-layer PCB 100 may correspond to the first substrate 111.

(149) One end of the first shielding via 710 may be physically connected to the (1a)th shielding pad 710a. The other end of the first shielding via 710 may be physically connected to the (1c)th shielding pad 710c.

(150) The (1a)th shielding pad 710a may be disposed on the third substrate 131. The (1a)th shielding pad 710a may be disposed on the first step of the first shielding via 710. The (1a)th shielding pad 710a may be disposed adjacent to the (1b)th via pad 310b in the third substrate 131. The diameter r7 of the first shielding via 710 may be determined based on at least one of an operation frequency of the antenna structure 10 and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure 10 is 140 GHz, the diameter r7 of the first shielding via 710 may be approximately 70 μm . For example, the diameter r8 of the (1a)th shielding pad 710a may be determined based on at least one of an operation frequency of the antenna structure 10 and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure 10 is 140 GHz, the diameter r8 of the (1a)th shielding pad 710a may be approximately 130 μm .

(151) The (1b)th shielding pad 710b may be disposed on the second step of the first shielding via 710. The (1b)th shielding pad 710b may be spaced apart by the predetermined height h3 from the (1a)th shielding pad 710a. The (1b)th shielding pad 710b may be adjacently disposed at a height substantially the same as the (1c)th via pad 310c. The diameter of the (1b)th shielding pad 710b may be equal to or similar to the diameter r8 of the (1a)th shielding pad 710a.

(152) The (1c)th shielding pad 710c may be disposed on the third step of the first shielding via 710. The (1c)th shielding pad 710c may be spaced apart by the predetermined height h4 from the (1b)th shielding pad 710b. The diameter of the (1c)th shielding pad 710c may be equal to or similar to the diameter r8 of the (1a)th shielding pad 710a.

(153) One end of the second shielding via 720 may be physically connected to the (2a)th shielding pad 720a. The other end of the second shielding via 720 may be physically connected to the (2c)th shielding pad 720c.

(154) The (2a)th shielding pad 720a may be disposed on the third substrate 131. The (2a)th shielding pad 720a may be disposed on the first step of the second shielding via 720. The (2a)th shielding pad 720a may be disposed adjacent to the (1b)th via pad 310b in the third substrate 131. The diameter of the second shielding via 720 may be equal to or similar to the diameter r7 of the first shielding via 710. The diameter of the (2a)th shielding pad 720a may be equal to or similar to the diameter r8 of the (1a)th shielding pad 710a.

(155) The (2b)th shielding pad 720b may be disposed on the second step of the second shielding via 720. The (2b)th shielding pad 720b may be spaced apart by the predetermined height h3 from the (2a)th shielding pad 720a. The (2b)th shielding pad 720b may be adjacently disposed at a height substantially the same as the (1c)th via pad 310c. The diameter of the (2b)th shielding pad 720b may be equal to or similar to the diameter r8 of the (1a)th shielding pad 710a.

(156) The (2c)th shielding pad **720c** may be disposed on the third step of the second shielding via **720**. The (2c)th shielding pad **720c** may be spaced apart by the predetermined height h_4 from the (2b)th shielding pad **720b**. The diameter of the (2c)th shielding pad **720c** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(157) The first shielding cover **771** may be spaced apart by the predetermined height h_4 from the (1c)th via pad **310c**. The first shielding cover **771** may be disposed at a height substantially the same as the first antenna patch **400**.

(158) One end of the third shielding via **730** may be physically connected to the (3a)th shielding pad **730a**. The other end of the third shielding via **730** may be physically connected to the (3c)th shielding pad **730c**. The diameter of the third shielding via **730** may be equal to or similar to the diameter r_7 of the first shielding via **710**. The diameter of the (3a)th shielding pad **730a** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(159) The (3a)th shielding pad **730a** may be disposed on the third substrate **131**. The (3a)th shielding pad **730a** may be disposed on the first step of the third shielding via **730**. The (3a)th shielding pad **730a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(160) The (3b)th shielding pad **730b** may be disposed on the second step of the third shielding via **730**. The (3b)th shielding pad **730b** may be spaced apart by the predetermined height h_3 from the (3a)th shielding pad **730a**. The (3b)th shielding pad **730b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**. The diameter of the (3b)th shielding pad **730b** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(161) The (3c)th shielding pad **730c** may be disposed on the third step of the third shielding via **730**. The (3c)th shielding pad **730c** may be spaced apart by the predetermined height h_4 from the (3b)th shielding pad **730b**. The diameter of the (3c)th shielding pad **730c** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(162) One end of the fourth shielding via **740** may be physically connected to the (4a)th shielding pad **740a**. The other end of the fourth shielding via **740** may be physically connected to the (4c)th shielding pad **740c**. The diameter of the fourth shielding via **740** may be equal to or similar to the diameter r_7 of the first shielding via **710**. The diameter of the (4a)th shielding pad **740a** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(163) The (4a)th shielding pad **740a** may be disposed on the third substrate **131**. The (4a)th shielding pad **740a** may be disposed on the first step of the fourth shielding via **740**. The (4a)th shielding pad **740a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(164) The (4b)th shielding pad **740b** may be disposed on the second step of the fourth shielding via **740**. The (4b)th shielding pad **740b** may be spaced apart by the predetermined height h_3 from the (4a)th shielding pad **740a**. The (4b)th shielding pad **740b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**. The diameter of the (4b)th shielding pad **740b** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(165) The (4c)th shielding pad **740c** may be disposed on the third step of the fourth shielding via **740**. The (4c)th shielding pad **740c** may be spaced apart by the predetermined height h_4 from the (4b)th shielding pad **740b**. The diameter of the (4c)th shielding pad **740c** may be equal to or similar to the diameter r_8 of the (1a)th shielding pad **710a**.

(166) The second shielding cover **772** may be spaced apart by the predetermined height h_4 from the (3c)th via pad **330c**. The second shielding cover **772** may be disposed at a height substantially the same as the probe feed.

(167) FIG. **17** is a perspective view showing an antenna structure including a third shielding structure **703** and a fourth shielding structure according to an embodiment of the disclosure.

(168) Referring to FIG. **17**, identically to the antenna structure **10** in FIGS. **9** to **12**, the antenna structure **10** may include a multi-layer PCB **100**, a first feed line **210**, a second feed line **220**, a third feed line **230**, a fourth feed line **240**, a first impedance matching structure **250**, a second impedance matching structure **260**, a first via **310**, a second via **320**, a third via **330**, a fourth via

340, a (1a)th via pad **310a**, a (1b)th via pad **310b**, a (1c)th via pad **310c**, a (2a)th via pad **320a**, a (2b)th via pad **320b**, a (3a)th via pad **330a**, a (3b)th via pad **330b**, a (3c)th via pad **330c**, a (4a)th via pad **340a**, a (4b)th via pad **340b**, a first antenna patch **400**, and a second antenna patch **500**.

(169) The antenna structure **10** may include a third shielding structure **703** and a fourth shielding structure **704**. For example, the third shielding structure **703** may be disposed to surround a part of each of the first via **310**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the second feed line **220**. The fourth shielding structure **704** may be disposed to surround a part of each of the third via **330**, the (3b)th via pad **330b**, the (3c)th via pad **330c**, and the fourth feed line **240**.

(170) The third shielding structure **703** may include a fifth shielding via **750**, a (5a)th shielding pad **750a**, a (5b)th shielding pad **750b**, a (5c)th shielding pad **750c**, and a third shielding cover **773**. The fourth shielding structure **704** may include a sixth shielding via **760**, a (6a)th shielding pad **760a**, a (6b)th shielding pad **760b**, a (6c)th shielding pad **760c**, and a fourth shielding cover **774**.

(171) FIG. **18** is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure.

(172) Referring to FIG. **18**, the third shielding structure **703** and the fourth shielding structure **704** may be arranged to face each other on the x-y plane.

(173) The third shielding structure **703** may be disposed substantially parallel to a first side surface of the first antenna patch **400**. The third shielding cover **773** may be disposed substantially parallel to a first side surface of the first antenna patch **400**. The third shielding structure **703** may be spaced apart by a predetermined distance w_{19} from the first side surface of the first antenna patch **400**. The third shielding cover **773** may be spaced apart by a predetermined distance w_{19} from the first side surface of the first antenna patch **400**. For example, the predetermined distance w_{19} may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_{19} may be approximately 105 μm .

(174) One end of the third shielding cover **773** may be physically coupled to the (5c)th shielding pad **750c**. The other end of the third shielding cover **773** may be disposed on the upper end of the (1c)th via pad **310c**. A horizontal length w_{22} of the third shielding cover **773** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the horizontal length w_{22} of the third shielding cover **773** may be approximately 300 μm . A vertical length d_4 of the third shielding cover **773** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the vertical length d_4 of the third shielding cover **773** may be approximately 130 μm .

(175) The (5c)th shielding pad **750c** may be spaced apart by a predetermined distance w_{23} from (1c)th via pad **310c** on the x-y plane. For example, the predetermined distance w_{23} may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance w_{23} may be approximately 40 μm . The (5a)th shielding pad **750a**, the (5b)th shielding pad **750b**, and the (5c)th shielding pad **750c** may be arranged to overlap each other on the z-axis.

(176) The fourth shielding structure **704** may be disposed substantially parallel to a second side surface of the first antenna patch **400**. The fourth shielding cover **774** may be disposed substantially parallel to a second side surface of the first antenna patch **400**. The fourth shielding structure **704** may be spaced apart by a predetermined distance w_{19} from the second side surface of the first antenna patch **400**. The fourth shielding cover **774** may be spaced apart by a predetermined distance from the second side surface of the first antenna patch **400**. For example, the spaced

distance between the fourth shielding structure **704** and the second side surface of the first antenna patch **400** may be equal to as or similar to the spaced distance w_{19} between the third shielding structure **703** and the first side surface of the first antenna patch **400**. For example, the spaced distance between the fourth shielding cover **774** and the second side surface of the first antenna patch **400** may be equal to as or similar to the spaced distance w_{19} between the third shielding cover **773** and the first side surface of the first antenna patch **400**.

(177) One end of the fourth shielding cover **774** may be physically coupled to the (6c)th shielding pad **760c**. The other end of the fourth shielding cover **774** may be disposed on the upper end of the (3c)th via pad **330c**. The horizontal length of the fourth shielding cover **774** may be equal to or similar to the horizontal length w_{22} of the third shielding cover **773**.

(178) The (6c)th shielding pad **760c** may be spaced apart by a predetermined distance from the (3c)th via pad **330c** on the x-y plane. For example, the spaced distance between the (6c)th shielding pad **760c** and the (3c)th via pad **330c** may be equal to or similar to the spaced distance w_{23} between the (5c)th shielding pad **750c** and the (1c)th via pad **310c**.

(179) The (6a)th shielding pad **760a**, the (6b)th shielding pad **760b**, and the (6c)th shielding pad **760c** may be arranged to overlap each other on the z-axis.

(180) FIG. **19** is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure.

(181) FIG. **20** is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(182) Referring to FIGS. **19** and **20**, the multi-layer PCB **100** may include a first layer to a sixth layer. For example, the first layer of the multi-layer PCB **100** may correspond to the second antenna patch **500**. The second layer of the multi-layer PCB **100** may correspond to at least one of the first antenna patch **400**, the third shielding cover **773**, and the fourth shielding cover **774**. The third layer of the multi-layer PCB **100** may correspond to at least one of the (3a)th via pad **330a**, the (4a)th via pad **340a**, the (5b)th shielding pad **750b**, and the (6b)th shielding pad **760b**. The fourth layer of the multi-layer PCB **100** may correspond to the third substrate **131**. The fifth layer of the multi-layer PCB **100** may correspond to the second substrate **121**. The sixth layer of the multi-layer PCB **100** may correspond to the first substrate **111**.

(183) One end of the fifth shielding via **750** may be physically connected to the (5c)th shielding pad **750c**. The other end of the fifth shielding via **750** may be spaced apart by the predetermined height h_4 from the (1c)th via pad **310c**.

(184) The (5a)th shielding pad **750a** may be disposed on the third substrate **131**. The (5a)th shielding pad **750a** may be disposed on the first step of the fifth shielding via **750**. The (5a)th shielding pad **750a** may be disposed adjacent to the (1b)th via pad **310b** in the third substrate **131**.

(185) A diameter r_9 of the fifth shielding via **750** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r_9 of the fifth shielding via **750** may be approximately 70 μm . For example, the diameter r_{10} of the (5a)th shielding pad **750a** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r_{10} of the (5a)th shielding pad **750a** may be approximately 130 μm .

(186) The (5b)th shielding pad **750b** may be disposed on the second step of the fifth shielding via **750**. The (5b)th shielding pad **750b** may be spaced apart by the predetermined height h_3 from the (5a)th shielding pad **750a**. The (5b)th shielding pad **750b** may be adjacently disposed at a height substantially the same as the (1c)th via pad **310c**. The diameter of the (5b)th shielding pad **750b** may be equal to or similar to the diameter r_{10} of the (5a)th shielding pad **750a**.

(187) The (5c)th shielding pad **750c** may be disposed on the third step of the fifth shielding via

750. The (5c)th shielding pad **750c** may be spaced apart by the predetermined height h_4 from the (5b)th shielding pad **750b**. The diameter of the (5c)th shielding pad **750c** may be equal to or similar to the diameter r_0 of the (5a)th shielding pad **750a**.

(188) The third shielding cover **773** may be spaced apart by the predetermined height h_4 from the (1c)th via pad **310c**. The third shielding cover **773** may be disposed at a height substantially the same as the probe feed.

(189) One end of the sixth shielding via **760** may be physically connected to the (6c)th shielding pad **760c**. The other end of the sixth shielding via **760** may be spaced apart by the predetermined height h_4 from the (3c)th via pad **330c**.

(190) The (6a)th shielding pad **760a** may be disposed on the third substrate **131**. The (6a)th shielding pad **760a** may be disposed on the first step of the sixth shielding via **760**. The (6a)th shielding pad **760a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(191) The diameter of the sixth shielding via **760** may be equal to or similar to the diameter r_9 of the fifth shielding via **750**. The diameter of the (6a)th shielding pad **760a** may be equal to or similar to the diameter r_{10} of the (5a)th shielding pad **750a**.

(192) The (6b)th shielding pad **760b** may be disposed on the second step of the sixth shielding via **760**. The (6b)th shielding pad **760b** may be spaced apart by the predetermined height h_3 from the (6a)th shielding pad **760a**. The (6b)th shielding pad **760b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**. The diameter of the (6b)th shielding pad **760b** may be equal to or similar to the diameter r_{10} of the (5a)th shielding pad **750a**.

(193) The (6c)th shielding pad **760c** may be disposed on the third step of the sixth shielding via **760**. The (6c)th shielding pad **760c** may be spaced apart by the predetermined height h_4 from the (6b)th shielding pad **760b**. The diameter of the (6c)th shielding pad **760c** may be equal to or similar to the diameter r_0 of the (5a)th shielding pad **750a**.

(194) The fourth shielding cover **774** may be spaced apart by the predetermined height h_4 from the (3c)th via pad **330c**. The fourth shielding cover **774** may be disposed at a height substantially the same as the first antenna patch **400**.

(195) FIG. **21** is a perspective view showing an antenna structure including a fifth shielding structure and a sixth shielding structure according to an embodiment of the disclosure.

(196) Referring to FIG. **21**, identically to the antenna structure **10** in FIGS. **9** to **12**, the antenna structure **10** may include a multi-layer PCB **100**, a first feed line **210**, a second feed line **220**, a third feed line **230**, a fourth feed line **240**, a first impedance matching structure **250**, a second impedance matching structure **260**, a first via **310**, a second via **320**, a third via **330**, a fourth via **340**, a (1a)th via pad **310a**, a (1b)th via pad **310b**, a (1c)th via pad **310c**, a (2a)th via pad **320a**, a (2b)th via pad **320b**, a (3a)th via pad **330a**, a (3b)th via pad **330b**, a (3c)th via pad **330c**, a (4a)th via pad **340a**, a (4b)th via pad **340b**, a first antenna patch **400**, and a second antenna patch **500**.

(197) The antenna structure **10** may include a fifth shielding structure and a sixth shielding structure. For example, the fifth shielding structure may be disposed to surround a part of each of the first via **310**, the (1b)th via pad **310b**, the (1c)th via pad **310c**, and the second feed line **220**. The sixth shielding structure may be disposed to surround a part of each of the third via **330**, the (3b)th via pad **330b**, the (3c)th via pad **330c**, and the fourth feed line **240**.

(198) The fifth shielding structure may include a first shielding via **710**, a second shielding via **720**, a fifth shielding via **750**, a (1a)th shielding pad **710a**, a (1b)th shielding pad **710b**, a (1c)th shielding pad **710c**, a (2a)th shielding pad **720a**, a (2b)th shielding pad **720b**, a (2c)th shielding pad **720c**, a (5a)th shielding pad **750a**, a (5b)th shielding pad **750b**, a (5c)th shielding pad **750c**, and a fifth shielding cover **775**.

(199) The sixth shielding structure may include a third shielding via **730**, a fourth shielding via **740**, a sixth shielding via **760**, a (3a)th shielding pad **730a**, a (3b)th shielding pad **730b**, a (3c)th shielding pad **730c**, a (4a)th shielding pad **740a**, a (4b)th shielding pad **740b**, a (4c)th shielding pad **740c**, a (6a)th shielding pad **760a**, a (6b)th shielding pad **760b**, a (6c)th shielding pad **760c**, and a

sixth shielding cover **776**.

(200) FIG. **22** is a plan view showing an antenna structure with reference to the x-y axis according to an embodiment of the disclosure.

(201) Referring to FIG. **22**, the fifth shielding structure and the sixth shielding structure may be arranged to face each other on the x-y plane.

(202) The fifth shielding structure may be disposed substantially parallel to a first side surface of the first antenna patch **400**. The fifth shielding cover **775** may be disposed substantially parallel to a first side surface of the first antenna patch **400**. The fifth shielding structure may be spaced apart by a predetermined distance **w19** from the first side surface of the first antenna patch **400**. The fifth shielding cover **775** may be spaced apart by a predetermined distance **w19** from the first side surface of the first antenna patch **400**. The predetermined distance **w19** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w19** may be approximately 105 μm .

(203) A first step of the fifth shielding cover **775** may be physically coupled to the (1c)th shielding pad **710c**. A second step of the fifth shielding cover **775** may be physically coupled to the (2c)th shielding pad **720c**. A third step of the fifth shielding cover **775** may be physically coupled to the (5c)th shielding pad **750c**.

(204) The (1c)th shielding pad **710c** may be spaced apart by a predetermined distance **w21** from (1c)th via pad **310c** on the x-y plane. For example, the predetermined distance **w21** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w21** may be approximately 40 μm . The (1a)th shielding pad **710a**, the (1b)th shielding pad **710b**, and the (1c)th shielding pad **710c** may be arranged to overlap each other on the z-axis.

(205) The (2c)th shielding pad **720c** may be spaced apart by a predetermined distance from the (1c)th via pad **310c** on the x-y plane. For example, the spaced distance between the (2c)th shielding pad **720c** and the (1c)th via pad **310c** may be equal to or similar to the spaced distance **w21** between the (1c)th shielding pad **710c** and the (1c)th via pad **310c**. The (2c)th shielding pad **720c** may be disposed to face the (1c)th via pad **710c** with reference to the (1c)th via pad **310c**. The (2a)th shielding pad **720a**, the (2b)th shielding pad **720b**, and the (2c)th shielding pad **720c** may be arranged to overlap each other on the z-axis.

(206) The (5c)th shielding pad **750c** may be spaced apart by a predetermined distance **w23** from (1c)th via pad **310c** on the x-y plane. For example, the predetermined distance **w23** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the predetermined distance **w23** may be approximately 40 μm . The (5a)th shielding pad **750a**, the (5b)th shielding pad **750b**, and the (5c)th shielding pad **750c** may be arranged to overlap each other on the z-axis.

(207) The sixth shielding structure may be disposed substantially parallel to a second side surface of the probe feed. The sixth shielding cover **776** may be disposed substantially parallel to the second side surface of the probe feed. The sixth shielding structure may be spaced apart by a predetermined distance from the second side surface of the probe feed. The sixth shielding cover **776** may be spaced apart by a predetermined distance from the second side surface of the probe feed.

(208) A first step of the sixth shielding cover **776** may be physically coupled to the (3c)th shielding pad **730c**. A second step of the sixth shielding cover **776** may be physically coupled to the (4c)th shielding pad **740c**. A third step of the sixth shielding cover **776** may be physically coupled to the (6c)th shielding pad **760c**.

(209) The (3c)th shielding pad **730c** may be spaced apart by a predetermined distance from the (3c)th via pad **330c** on the x-y plane. For example, the spaced distance between the (3c)th shielding pad **730c** and the (3c)th via pad **330c** may be equal to or similar to the spaced distance w21 between the (1c)th shielding pad **710c** and the (1c)th via pad **310c**. The (3a)th shielding pad **730a**, the (3b)th shielding pad **730b**, and the (3c)th shielding pad **730c** may be arranged to overlap each other on the z-axis.

(210) The (4c)th shielding pad **740c** may be spaced apart by a predetermined distance from the (3c)th via pad **330c** on the x-y plane. For example, the spaced distance between the (4c)th shielding pad **740c** and the (3c)th via pad **330c** may be equal to or similar to the spaced distance w21 between the (1c)th shielding pad **710c** and the (1c)th via pad **310c**. The (4c)th shielding pad **740c** may be disposed to face the (3c)th via pad **730c** with reference to the (3c)th via pad **330c**. The (4a)th shielding pad **740a**, the (4b)th shielding pad **740b**, and the (4c)th shielding pad **740c** may be arranged to overlap each other on the z-axis.

(211) The (6c)th shielding pad **760c** may be spaced apart by a predetermined distance from (3c)th via pad **330c** on the x-y plane. For example, the spaced distance between the (6c)th shielding pad **760c** and the (3c)th via pad **330c** may be equal to or similar to the spaced distance w21 between the (1c)th shielding pad **710c** and the (1c)th via pad **310c**.

(212) The (6a)th shielding pad **760a**, the (6b)th shielding pad **760b**, and the (6c)th shielding pad **760c** may be arranged to overlap each other on the z-axis.

(213) FIG. **23** is a cross-sectional view showing an antenna structure with reference to the y-z axis according to an embodiment of the disclosure.

(214) FIG. **24** is a cross-sectional view showing an antenna structure with reference to the x-z axis according to an embodiment of the disclosure.

(215) Referring to FIGS. **23** and **24**, the multi-layer PCB **100** may include a first layer to a sixth layer. For example, the first layer of the multi-layer PCB **100** may correspond to the second antenna patch **500**. The second layer of the multi-layer PCB **100** may correspond to at least one of the first antenna patch **400**, the fifth shielding cover **775**, and the sixth shielding cover **776**. The third layer of the multi-layer PCB **100** may correspond to at least one of the (3a)th via pad **330a**, the (4a)th via pad **340a**, the (1b)th shielding pad **710b**, the (2b)th shielding pad **720b**, the (3b)th shielding pad **730b**, the (4b)th shielding pad **740b**, the (5b)th shielding pad **750b**, and the (6b)th shielding pad **760b**. The fourth layer of the multi-layer PCB **100** may correspond to the third substrate **131**. The fifth layer of the multi-layer PCB **100** may correspond to the second substrate **121**. The sixth layer of the multi-layer PCB **100** may correspond to the first substrate **111**.

(216) One end of the first shielding via **710** may be physically connected to the (1a)th shielding pad **710a**. The other end of the first shielding via **710** may be physically connected to the (1c)th shielding pad **710c**.

(217) The (1a)th shielding pad **710a** may be disposed on the third substrate **131**. The (1a)th shielding pad **710a** may be disposed on the first step of the first shielding via **710**. The (1a)th shielding pad **710a** may be disposed adjacent to the (1b)th via pad **310b** in the third substrate **131**. The diameter r7 of the first shielding via **710** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r7 of the first shielding via **710** may be approximately 70 μm . The diameter r8 of the (1a)th shielding pad **710a** may be determined based on at least one of an operation frequency of the antenna structure **10** and a minimum limitation in the process at the time of manufacturing thereof. For example, in case that an operation frequency of the antenna structure **10** is 140 GHz, the diameter r8 of the (1a)th shielding pad **710a** may be approximately 130 μm .

(218) The (1b)th shielding pad **710b** may be disposed on the second step of the first shielding via **710**. The (1b)th shielding pad **710b** may be spaced apart by the predetermined height h3 from the (1a)th shielding pad **710a**. The (1b)th shielding pad **710b** may be adjacently disposed at a height

substantially the same as the (1c)th via pad **310c**. The diameter of the (1b)th shielding pad **710b** may be equal to or similar to the diameter r8 of the (1a)th shielding pad **710a**.

(219) The (1c)th shielding pad **710c** may be disposed on the third step of the first shielding via **710**. The (1c)th shielding pad **710c** may be spaced apart by the predetermined height h4 from the (1b)th shielding pad **710b**. The diameter of the (1c)th shielding pad **710c** may be equal to or similar to the diameter r8 of the (1a)th shielding pad **710a**.

(220) One end of the second shielding via **720** may be physically connected to the (2a)th shielding pad **720a**. The other end of the second shielding via **720** may be physically connected to the (2c)th shielding pad **720c**.

(221) The (2a)th shielding pad **720a** may be disposed on the third substrate **131**. The (2a)th shielding pad **720a** may be disposed on the first step of the second shielding via **720**. The (2a)th shielding pad **720a** may be disposed adjacent to the (1b)th via pad **310b** in the third substrate **131**. The diameter of the second shielding via **720** may be equal to or similar to the diameter r7 of the first shielding via **710**. The diameter of the (2a)th shielding pad **720a** may be equal to or similar to the diameter r8 of the (1a)th shielding pad **710a**.

(222) The (2b)th shielding pad **720b** may be disposed on the second step of the second shielding via **720**. The (2b)th shielding pad **720b** may be spaced apart by the predetermined height h3 from the (2a)th shielding pad **720a**. The (2b)th shielding pad **720b** may be adjacently disposed at a height substantially the same as the (1c)th via pad **310c**. The diameter of the (2b)th shielding pad **720b** may be equal to or similar to the diameter r8 of the (1a)th shielding pad **710a**.

(223) The (2c)th shielding pad **720c** may be disposed on the third step of the second shielding via **720**. The (2c)th shielding pad **720c** may be spaced apart by the predetermined height h4 from the (2b)th shielding pad **720b**.

(224) One end of the fifth shielding via **750** may be physically connected to the (5c)th shielding pad **750c**. The other end of the fifth shielding via **750** may be spaced apart by the predetermined height h4 from the (1c)th via pad **310c**.

(225) The (5a)th shielding pad **750a** may be disposed on the third substrate **131**. The (5a)th shielding pad **750a** may be disposed on the first step of the fifth shielding via **750**. The (5a)th shielding pad **750a** may be disposed adjacent to the (1b)th via pad **310b** in the third substrate **131**.

(226) The (5b)th shielding pad **750b** may be disposed on the second step of the fifth shielding via **750**. The (5b)th shielding pad **750b** may be spaced apart by the predetermined height h3 from the (5a)th shielding pad **750a**. The (5b)th shielding pad **750b** may be adjacently disposed at a height substantially the same as the (1c)th via pad **310c**.

(227) The (5c)th shielding pad **750c** may be disposed on the third step of the fifth shielding via **750**. The (5c)th shielding pad **750c** may be spaced apart by the predetermined height h4 from the (5b)th shielding pad **750b**.

(228) The fifth shielding cover **775** may be spaced apart by the predetermined height h4 from the (1c)th via pad **310c**. The fifth shielding cover **775** may be disposed at a height substantially the same as the first antenna patch **400**.

(229) One end of the third shielding via **730** may be physically connected to the (3a)th shielding pad **730a**. The other end of the third shielding via **730** may be physically connected to the (3c)th shielding pad **730c**.

(230) The (3a)th shielding pad **730a** may be disposed on the third substrate **131**. The (3a)th shielding pad **730a** may be disposed on the first step of the third shielding via **730**. The (3a)th shielding pad **730a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(231) The (3b)th shielding pad **730b** may be disposed on the second step of the third shielding via **730**. The (3b)th shielding pad **730b** may be spaced apart by the predetermined height h3 from the (3a)th shielding pad **730a**. The (3b)th shielding pad **730b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**.

(232) The (3c)th shielding pad **730c** may be disposed on the third step of the third shielding via

730. The (3c)th shielding pad **730c** may be spaced apart by the predetermined height h_4 from the (3b)th shielding pad **730b**.

(233) One end of the fourth shielding via **740** may be physically connected to the (4a)th shielding pad **740a**. The other end of the fourth shielding via **740** may be physically connected to the (4c)th shielding pad **740c**.

(234) The (4a)th shielding pad **740a** may be disposed on the third substrate **131**. The (4a)th shielding pad **740a** may be disposed on the first step of the fourth shielding via **740**. The (4a)th shielding pad **740a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(235) The (4b)th shielding pad **740b** may be disposed on the second step of the fourth shielding via **740**. The (4b)th shielding pad **740b** may be spaced apart by the predetermined height h_3 from the (4a)th shielding pad **740a**. The (4b)th shielding pad **740b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**.

(236) The (4c)th shielding pad **740c** may be disposed on the third step of the fourth shielding via **740**. The (4c)th shielding pad **740c** may be spaced apart by the predetermined height h_4 from the (4b)th shielding pad **740b**.

(237) One end of the sixth shielding via **760** may be physically connected to the (6c)th shielding pad **760c**. The other end of the sixth shielding via **760** may be spaced apart by the predetermined height h_4 from the (3c)th via pad **330c**.

(238) The (6a)th shielding pad **760a** may be disposed on the third substrate **131**. The (6a)th shielding pad **760a** may be disposed on the first step of the sixth shielding via **760**. The (6a)th shielding pad **760a** may be disposed adjacent to the (3b)th via pad **330b** in the third substrate **131**.

(239) The (6b)th shielding pad **760b** may be disposed on the second step of the sixth shielding via **760**. The (6b)th shielding pad **760b** may be spaced apart by the predetermined height h_3 from the (6a)th shielding pad **760a**. The (6b)th shielding pad **760b** may be adjacently disposed at a height substantially the same as the (3c)th via pad **330c**.

(240) The (6c)th shielding pad **760c** may be disposed on the third step of the sixth shielding via **760**. The (6c)th shielding pad **760c** may be spaced apart by the predetermined height h_4 from the (6b)th shielding pad **760b**.

(241) The sixth shielding cover **776** may be spaced apart by the predetermined height h_4 from the (3c)th via pad **330c**. The sixth shielding cover **776** may be disposed at a height substantially the same as the first antenna patch **400**.

(242) FIG. **25** is a graph showing an isolation characteristic between antenna ports and of the antenna structure according to an embodiment of the disclosure.

(243) Referring to FIG. **25**, an isolation characteristic S_{21} (S-parameter, dB) between the first antenna port **210** and the second antenna port **220** according to the antenna structure **10** in FIGS. **9** to **12** may be shown as in a first graph **2501**. For example, in case that an operation frequency of the antenna structure **10** in FIGS. **9** to **12** is 140 GHz, a value of the isolation characteristic S_{21} between the first antenna port **210** and the second antenna port **220** may be -12.63 dB.

(244) An isolation characteristic S_{21} (S-parameter, dB) between the first antenna port **210** and the second antenna port **220** according to the antenna structure **10** in FIGS. **13** to **16** may be shown as in a second graph **2502**. For example, in case that an operation frequency of the antenna structure **10** in FIGS. **13** to **16** is 140 GHz, a value of the isolation characteristic S_{21} between the first antenna port **210** and the second antenna port **220** may be -20.71 dB.

(245) An isolation characteristic S_{21} (S-parameter, dB) between the first antenna port **210** and the second antenna port **220** according to the antenna structure **10** in FIGS. **17** to **20** may be shown as in a third graph **2503**. For example, in case that an operation frequency of the antenna structure **10** in FIGS. **17** to **20** is 140 GHz, a value of the isolation characteristic S_{21} between the first antenna port **210** and the second antenna port **220** may be approximately -17.63 dB.

(246) An isolation characteristic S_{21} (S-parameter, dB) between the first antenna port **210** and the second antenna port **220** according to the antenna structure **10** in FIGS. **21** to **24** may be shown as

in a fourth graph 2504. For example, in case that an operation frequency of the antenna structure 10 in FIGS. 21 to 24 is 140 GHz, a value of the isolation characteristic S21 between the first antenna port 210 and the second antenna port 220 may be approximately -21.17 dB.

(247) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An antenna module in a wireless communication system, the antenna module comprising: a multi-layer printed circuit board (PCB) comprising a first to a sixth layer; a first feed line disposed in a first area of one side surface of the fifth layer of the multi-layer PCB; a second feed line disposed in a second area of the one side surface of the fifth layer to face the first feed line; a first via extending through the first area of the fifth layer and a first area of the fourth layer of the multi-layer PCB, and having one end electrically connected to the first feed line and the other end extending by a first height in a vertical direction from the first area of the fourth layer; a second via extending through the second area of the fifth layer and a second area of the fourth layer while facing the first via, and having one end electrically connected to the second feed line and the other end extending by the first height in a vertical direction from the second area of the fourth layer; a third feed line having one end electrically connected to the other end of the first via, and the other end extending horizontally in a first direction from the other end of the first via; a fourth feed line facing the third feed line, having one end electrically connected to the other end of the second via, and extending horizontally in a second direction from the other end of the second via; a third via having one end electrically connected to the other end of the third feed line, and the other end configured to extend through a fourth area of the third layer and extend by a second height in a vertical direction; a fourth via facing the third via, and having one end electrically connected to the other end of the fourth feed line and the other end configured to extend through the fourth area of the fourth layer and extend by the second height in a vertical direction; a first antenna patch having a lower surface electrically connected to the other end of the third via and the other end of the fourth via; and a second antenna patch spaced apart by a predetermined third height from an upper surface of the first antenna patch, wherein a spaced distance between the first via and the second via is greater than a spaced distance between the third via and the fourth via.

2. The antenna module of claim 1, wherein the first via comprises: a (1a)th via pad disposed in the first area of the fifth layer and physically coupled to one end of the first via; a (1b)th via pad disposed in the first area of the fourth layer and physically coupled to a middle step of the first via; and a (1c)th via pad physically connected to the other end of the first via.

3. The antenna module of claim 1, wherein the second via comprises: a (2a)th via pad disposed in the second area of the fifth layer and physically coupled to one end of the second via; a (3b)th via pad disposed in the second area of the fifth layer and physically coupled to a middle step of the second via; and a (3c)th via pad physically connected to the other end of the second via.

4. The antenna module of claim 1, wherein the third via comprises: a (3a)th via pad physically coupled to one end of the third via; and a (3b)th via pad physically coupled to the other end of the third via and embedded in a lower surface of a probe pad.

5. The antenna module of claim 1, wherein the fourth via comprises: a (4a)th via pad physically coupled to one end of the fourth via; and a (4b)th via pad physically coupled to the other end of the fourth via and embedded in a second area of a lower surface of a probe pad.

6. The antenna module of claim 1, wherein the first feed line and the second feed line are electrically connected to a radio frequency integrated circuit (RFIC).

7. The antenna module of claim 6, wherein a first signal output from the RFIC is emitted through

the second antenna patch via the first feed line, the first via, the third feed line, the third via, and a first area of the first antenna patch.

8. The antenna module of claim 6, wherein a second signal output from the RFIC is emitted through the second antenna patch via the second feed line, the second via, the fourth feed line, the fourth via, and a second area of the first antenna patch.

9. The antenna module of claim 1, further comprising: a first shielding structure configured to surround a first side surface, a second side surface, and an upper surface of the first via; and a second shielding structure configured to surround a first side surface, a second side surface, and an upper surface of the third via, wherein the first shielding structure and the second shielding structure are arranged symmetrically with each other.

10. The antenna module of claim 9, wherein the first shielding structure comprises: a first shielding via spaced apart by a predetermined distance from the first side surface of the first via; a second shielding via spaced apart by a predetermined distance from the second side surface of the first via; and a first shielding cover configured to extend from one end of the first shielding via to one end of the second shielding via and spaced apart by a predetermined distance from the upper surface of the first via.

11. The antenna module of claim 10, wherein the first shielding via is configured to extend from the third layer to a height which is substantially equal to the first antenna patch, and wherein the second shielding via is configured to extend from the third layer to a height which is substantially equal to the first antenna patch.

12. The antenna module of claim 10, wherein the first shielding cover is disposed at a height which is substantially equal to the first antenna patch so as to be substantially parallel to a first side surface of the first antenna patch.

13. The antenna module of claim 10, wherein the second shielding structure comprises: a third shielding via spaced apart by a predetermined distance from the first side surface of the second via; a fourth shielding via spaced apart by a predetermined distance from the second side surface of the second via; and a second shielding cover configured to extend from one end of the third shielding via to one end of the fourth shielding via and spaced apart by a predetermined distance from the upper surface of the second via.

14. The antenna module of claim 13, wherein the third shielding via is configured to extend from the third layer to a height which is substantially equal to the first antenna patch, and wherein the fourth shielding via is configured to extend from the third layer to a height which is substantially equal to the first antenna patch.

15. The antenna module of claim 13, wherein the second shielding cover is disposed at a height which is substantially equal to the first antenna patch so as to be substantially parallel to the second side surface of the first antenna patch.

16. The antenna module of claim 13, further comprising: a third shielding structure configured to surround a third side surface of the first via; and a fourth shielding structure configured to surround a third side surface of the third via, wherein the third shielding structure and the fourth shielding structure are arranged symmetrically with each other.

17. The antenna module of claim 16, wherein the third shielding structure comprises: a fifth shielding via spaced apart by a predetermined distance from the third side surface of the first via; and a third shielding cover configured to extend from one end of the fifth shielding via to the first shielding structure.

18. The antenna module of claim 17, wherein the fifth shielding via is configured to extend from the third layer to a height which is substantially equal to the first antenna patch, and wherein the third shielding cover is disposed at a height which is substantially equal to the first antenna patch so as to be substantially perpendicular to a first side surface of the first antenna patch on a x-y plane.

19. The antenna module of claim 17, wherein the fourth shielding structure comprises: a sixth

shielding via spaced apart by a predetermined distance from the third side surface of the third via; and a fourth shielding cover configured to extend from one end of the sixth shielding via to the second shielding structure.

20. The antenna module of claim 19, wherein the sixth shielding via is configured to extend from the third layer to a height which is substantially equal to a probe pad, and wherein the fourth shielding cover is disposed at a height which is substantially equal to the probe pad so as to be substantially perpendicular to a second side surface of the probe pad on an x-y plane.
